Number Systems

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How are numeric data items actually stored in the computer memory?

How much space(memory locations) is allocated for each type of data?

- int, float, char etc.

How are characters and strings are stored in the memory?

Number System:: The Basics

We are accustomed to using the so called **decimal number system.**

- ○ Ten digits::0,1,2,3,4,5,6,7,8,9
- ○ Every digit position has a weight which is A power of 10.
- ○ Base or radix is 10.

Example:

$234 = 2*10^2 + 3*10^1 + 4*10^0$

$250.67 = 2*10^2 + 5*10^1 + 0*10^0 + 6*10^{-1} + 7*10^{-2}$

<u>Binary Number System</u>

Two digits:

- **0 and 1**
- **Every digit position has a weight which is A power of two.**
- **Base or radix is 2.**

Example:

$\mathbf{110 = 1*2^2 + 1*2^1 + 0*2^0}$

$101.01 = 1*2^2 + 0*2^1 + 1*2^0 + 0*2^{-1} + 1*2^{-2}$

Binary to Decimal conversion

Each digit position of a binary number has a weight.

- ○ **Some power of two**

A binary number:

$\mathbf{B = b_{n-1}b_{n-2}\ldots b_1b_0b_{-1}b_{-2}\ldots b_{-m}}$

Corresponding value in decimal:

$$\boxed{D = \sum_{i=-m}^{n-1} b_i \cdot 2^i}$$

Examples:

$$\overset{5\,4\,3\,2\,1\,0}{101011} \rightarrow 1\times2^5 + 0\times2^4 + 1\times2^3 + 0\times2^2 + 1\times2^1 + 1\times2^0$$
$$= 32 + 0 + 8 + 0 + 2 + 1$$
$$= 43_{10}$$

$$0.0101 \rightarrow 0\times2^{-1} + 1\times2^{-2} + 0\times2^{-3} + 0\times2^{-4}$$
$$= 0 + \frac{1}{4} + 0 + \frac{1}{16} = \frac{4+1}{16} = \frac{5}{16} = 0.3125$$

$$= 0 + \frac{1}{4} + 0 + \frac{1}{16} = \frac{4+1}{16} = \frac{5}{16} = 0.3125$$

$$0.0101_2 = 0.3125_{10}$$

$$\overset{2\,1\,0\;\,-1\,-2}{101.11_2} = 1\times2^2 + 0\times2^1 + 1\times2^0 + 1\times2^{-1} + 1\times2^{-2}$$

$$= 4 + 0 + 1 + 0.5 + 0.25$$

$$101.11_2 = 5.75_{10}$$

Decimal-to-Binary Conversion

Consider the integer and fractional parts separately

For integer part,

- Repeatedly divide the given number by two and go on accumulating the remainders until the number becomes 0.
- Arrange the reminders in reverse order.

For the fractional part:

- Repeatedly multiply the given fraction by two.
  - Accumulate the integer part (0, 1).
  - If the integer part is one, chop it off.
- Arrange the integer part in order they are obtained.

Examples:

$$(239)_{10} = (\,?\,)_2$$

2	239	
2	119	1
2	59	1
2	29	1
2	14	1
2	7	0
2	3	1
2	1	1
	0	1

$$(239)_{10} = (11101111)_2$$

0 or 1

10
00
11
01

$2^2 \; 2^1 \; 2^0$

$\boxed{4}\,\boxed{2}\,\boxed{1}$ 0 to 7

0 1 1 3
1 0 1 5

5 4 3 2 1 0

$\boxed{32}\,\boxed{16}\,\boxed{8}\,\boxed{4}\,\boxed{2}\,\boxed{1}$ $\boxed{0 \text{ to } 63}$ $2^6 - 1$

$(1\;1\;1\;0\;0\;1)_2$ 57

$(1\;0\;1\;0\;0\;1)_2$ 41

$\boxed{2^n - 1}$

$2^{12} - 1$

11 10 9 8 7 6 5 4 3 2 1 0

$\boxed{2048}\,\boxed{1024}\,\boxed{512}\,\boxed{256}\,\boxed{128}\,\boxed{64}\,\boxed{32}\,\boxed{16}\,\boxed{8}\,\boxed{4}\,\boxed{2}\,\boxed{1}$ 0 to 4095

0.634

carry	
0	.634 × 2
1	.268 × 2
0	.536 × 2
1	.072 × 2

$$(0.634)_{10} = (0.01010\ldots)_2$$

0	536 × 2
1	072 × 2
0	144
⋮	⋮

$(37.625)_{10} = (\; ? \;)_2$

$37_{10} = 100101_2$

$37.625_{10} = 100101_2 + 0.101_2$
$= 100101.101_2$

32 16 8 4 2 1
1 0 0 1 0 1

0	625 × 2
1	250 × 2
0	500 × 2
1	000

Hexadecimal Number System
A compact way of representing binary numbers
16 different symbols(radix = 16)

4 digits. $2^3 = 8$

0000	→	0	←	0 0 0 0
0001	→	1	←	0 0 0 1
0010	→	2	←	0 0 1 0
0011	→	3	←	0 0 1 1
0100	→	4	←	0 1 0 0
0101	→	5	←	0 1 0 1
0110	→	6	←	0 1 1 0
0111	→	7	←	0 1 1 1
1000	→	8	←	1 0 0 0
1001	→	9	←	1 0 0 1
1010	→	10 or A	←	1 0 1 0
1011	→	11 or B	←	1 0 1 1
1100	→	12 or C	←	1 1 0 0
1101	→	13 or D	←	1 1 0 1
1110	→	14 or E	←	1 1 1 0
1111	→	15 or F	←	1 1 1 1

Binary to Hexadecimal Conversion

For the integer part:

- **Scan the binary number from right to left.**
- **Translate each group of four bits into the corresponding hexadecimal digit.**
  - **Add leading zeros if necessary.**

For the fractional part:

- **Scan the binary number from left to right.**
- **Translate each group of four bits into the corresponding hexadecimal digit.**
  - **Add trailing zeros if necessary.**

$(0101\ 1010\ 0011)_2 = (5A3)_{16}$

$(0101\,1010\,0011)_2 = (5A3)_{16}$

5 10 3

A

$(0.10000100)_2 = (0.84)_{16}$

8 4

$(0101.01011110)_2 = (5.5E)_{16}$

5 5 14 E

Hexadecimal to binary Conversion

Translate every hexadecimal digit into its 4 bit binary equivalent.

Examples:

$(3A5) = (11\,1010\,0101)_2$

$(12.3D)_{16} = (0001\,0010.0011\,1101)_2$

8 4 2 1
0 1 0 1
1 0 1 0
1 1 0 1
0 0 1 1
0 0 1 0

Octal Number System
A compact way of representing binary numbers
8 different symbols(radix = 8)

000 → 0
001 → 1
010 → 2
011 → 3
100 → 4
101 → 5
110 → 6
111 → 7

Binary to Octal Conversion

For the integer part:

- Scan the binary number from right to left.
- Translate each group of three bits into the corresponding hexadecimal digit.
    - Add leading zeros if necessary.

For the fractional part:

- Scan the binary number from left to right.
- Translate each group of three bits into the corresponding hexadecimal digit.
    - Add trailing zeros if necessary.

- **Translate each group of three bits into the corresponding hexadecimal digit.**
  - **Add trailing zeros if necessary.**

$$(00|1|010|101|101)_2 = (1255)_8$$

$$(.010|101|110)_2 = (0.256)_8$$

Octal to binary Conversion

Translate every hexadecimal digit into its 3 bit binary equivalent.

$$(437)_8 = (100\ 011\ 111)_2$$

$$(53.36)_8 = (101\ 011.011\ 110)_2$$

WAPJ to convert from one number system to another number system. Your program will generate a menu:
Press 1 Convert from decimal to binary
Press 2 Convert from binary to decimal
Press 3 Convert from decimal to hexadecimal
Press 4 Convert from hexadecimal to decimal
Press 5 Convert from hexadecimal to binary
Press 6 Convert from binary to hexadecimal
Press 7 Convert from binary to octal
Press 8 Convert from octal to binary
Press 9 Convert from decimal to octal
Press 10 Convert from octal to decimal
Press 11 Convert from octal to hexadecimal
Press 12 Convert from hexadecimal to octal
Enter your choice: